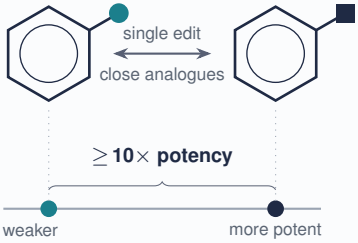
TELEMEDICON26 | PROJECT PITCH DECK

CliffForge

Relative-Affinity Molecular Analogue Prioritization and Activity-Cliff Analysis

Constrained analogue generation · relative-affinity scoring · uncertainty-aware ranking



an activity cliff (schematic)

Team Genesis

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Drug discovery: a short vocabulary

Ligand

A small molecule tested for interaction with a target protein.

Protein pocket

The three-dimensional region where the ligand sits and makes contacts.

Binding affinity

How strongly the ligand binds. Lower K_d means tighter binding.

Relative affinity

The change between two ligands against the same target, rather than either score alone.

Congeneric / within-series

Analogues that share a chemical core and differ through small local edits.

Activity cliff

Close analogues with a large difference in measured biological activity.

Our experimental label is pIC$_{50}$: a one-unit increase means a tenfold lower IC$_{50}$ in the assay. pIC$_{50}$ is a potency measure, not a direct binding-affinity measurement.

Sources: project teammate brief, pp. 1–3; activity-cliff definitions (Maggiora 2006; Stumpfe & Bajorath 2012).

A small edit can improve the measured response



Pocket-based hypothesis	The new contact may make B bind more favorably than A.
Improved assay effect	Illustrative outcome: B inhibits the target more strongly than A.

A $\geq$ 1-unit pIC$_{50}$ gain means $\geq$ 10$\times$ assay potency: a beneficial activity cliff.

A possible therapeutic improvement to test; no clinical response has been measured.

Conceptual interaction and positive-cliff example, not a measured pose or clinical result. Therapeutic benefit needs later testing; pIC$_{50}$ is assay potency (Boby et al. 2023; Stumpfe & Bajorath 2012).

Representative absolute activity and affinity predictors

Each architecture predicts one scalar endpoint for one candidate or drug–target pair.

MODEL	INPUT	ENCODER / HEAD	OUTPUT
ChemProp	Ligand graph only	Directed bond message passing + regression	Mpro pIC_{50} in our benchmark
DeepDTA	SMILES + protein sequence	Separate 1D CNNs; fused regression	Drug–target activity / affinity score
GraphDTA	Ligand graph + protein sequence	Graph neural net + protein CNN; regression	Drug–target activity / affinity score
Pafnucy	Posed 3D complex	Voxelized complex + 3D CNN; dense head	Affinity estimate from 3D structure

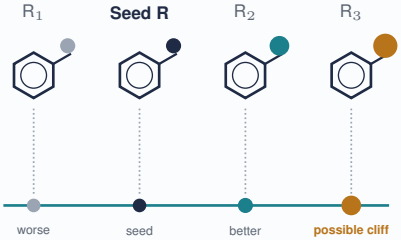
Close-pair comparison: predict A and B separately, then compute $\hat{y}_{\mathrm{B}} - \hat{y}_{\mathrm{A}}$.

ChemProp is the tested comparator here. Our pIC_{50} assay endpoint is not a direct K_d / K_i binding-affinity measurement.

Architecture sources: (Nguyen et al. 2021; Öztürk et al. 2018; Stepniewska-Dziubinska et al. 2018; Yang et al. 2019).

Our research focus

Predicting binding-affinity within series, with a focus on accurately predicting activity cliffs



Model task: rank candidate-minus-seed Δactivity

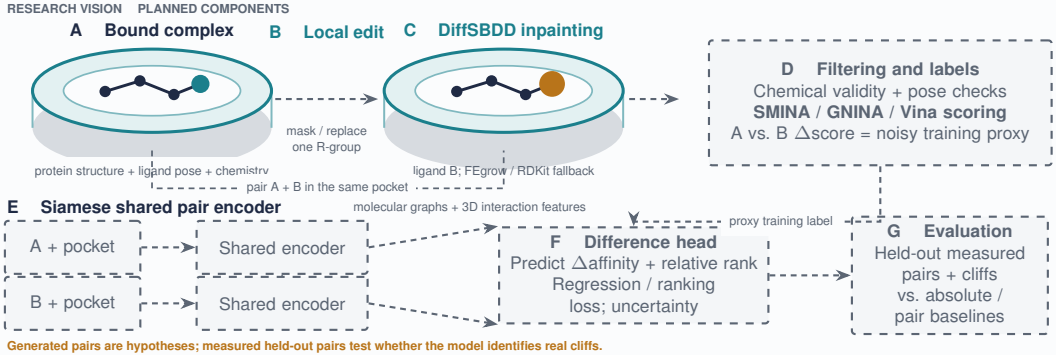
Why this matters to chemists

- Synthesis and assays take time; ranking close analogues helps choose the next experiment.
- A correct activity direction tells whether a small edit improves the current lead.
- A cliff alert draws attention to an unusually large gain or loss that similarity alone may hide.

The scientific test is held-out *measured* activity. Generated analogues and model scores are hypotheses until assayed.

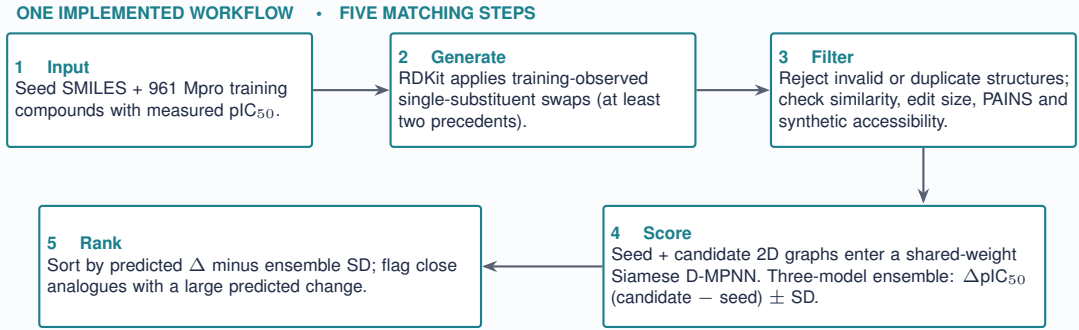
Lead-optimization framing: project teammate brief, pp. 1–3; matched-pair rationale (Griffen et al. 2011).

Proposed protein-aware research pipeline



Planned A–G sequence: CliffForge teammate brief, pp. 2–3. SMINA / GNINA / Vina supply proxy scores, not measured affinity. Slide 7 shows the current implementation.

Current prototype: a 2D relative-activity model



Current scope: 2D molecular graphs and rule-based edits; no protein pocket, stereochemistry, DiffSBDD or docking.

Implementation: `src/cliffforge/generate.py`, `relmodel.py`, `demo.py`; D-MPNN baseline (Yang et al. 2019).

Two Mpro datasets, two cliff definitions

COVID Moonshot noncovalent Mpro

1,102 molecules

One fixed molecule split: **961 train** / **141 test**. IC_{50} assay values become pIC_{50}; the prototype uses standardized 2D structures.

Cliff rule

Morgan-fingerprint Tanimoto ≥ 0.80 **and** measured $|\Delta pIC_{50}| \geq 1$ (at least $10\times$ potency).

46 cliff pairs overall. Among **44** close test–test pairs, only **5** are strict held-out cliffs.

Dablander cleaned Mpro MMPs

1,924 molecules

12,594 supplied matched molecular pairs (MMPs). Five molecule-level folds evaluate held-out compounds; supplied pair identities are retained.

Cliff rule

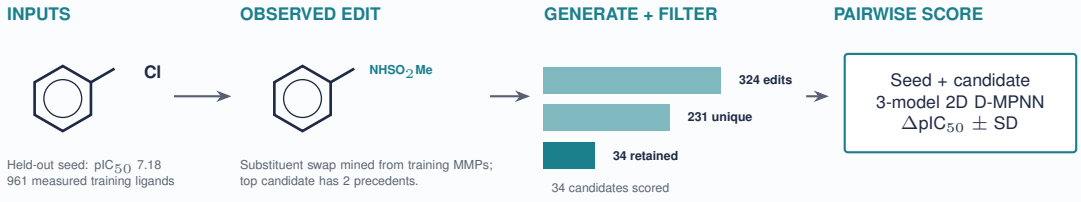
Measured $|\Delta pIC_{50}| \geq 2$ (at least $100\times$ potency). Noncliffs have $|\Delta| \leq 1$; intermediate pairs leave binary detection.

2,132 binary test–test MMPs contain **106** cliffs; **343** intermediate test–test pairs are excluded.

Both evaluations study Mpro activity. Their pair rules, cliff thresholds and splits differ, so the cliff rates are not pooled.

Sources: Moonshot (Boby et al. 2023); Dablander MMPs (Dablander et al. 2023); project protocol and result files.

One prototype run: from seed to ranked shortlist



Ranked output	EDIT	PREDICTED ΔpIC$_{50}$	SIMILARITY	CLIFF ALERT
#1	Cl $\rightarrow$ NHSO$_2$Me	**+0.18 $\pm$ 0.01**	0.90	No
#2	side-chain cyclopropyl edit	+0.15 $\pm$ 0.02	0.72	No
#3	sulfonamide $\rightarrow$ methoxy	+0.08 $\pm$ 0.05	0.75	No

Ranked by predicted Δ minus SD. Chemical motif is schematic; output values are from the real run. None was assayed or docked.

Source: `results/moonshot_mpro_tanimoto/metrics/shortlist.md` and `funnel.json`.

Retrospective comparison: CliffForge and ChemProp

DATASET	ACTIVITY-CLIFF RANKING	WITHIN-SERIES RANKING	LEAD OPTIMIZATION
Moonshot Mpro one split 1,102 molecules	AP: **0.73** vs. 0.35 Cliffs found: **2/5** vs. 0/5 44 close test–test pairs; fixed $\lvert\hat{\Delta}\rvert > 1.0$	Mean Spearman ρ: **0.581** vs. 0.192 12 held-out chemical series	Correct direction: **53/82** vs. 47/82 Retrospective proxy: 82/316 reachable pairs recovered. Direction gap inconclusive; no measured gain for generated candidates.
Dablander MMP five folds 1,924 molecules	AP: **0.391** vs. 0.137 Cliffs found: **20/106** vs. 0/106 2,132 binary test–test MMPs; $\lvert\hat{\Delta}\rvert > 1.5$. False positives: 16 vs. 0.	Mean Spearman ρ: **0.509** vs. 0.343 2,014 reference neighborhoods	Top 10% mean measured gain: **+1.116** vs. +1.002 pIC_{50} 2,026 proposals per model; gain versus ChemProp is inconclusive under adjusted inference.

Teal = CliffForge; dark text = ChemProp. AP is average precision. Different cliff definitions apply on the two datasets. Adjusted inference supports Dablander local ranking, while cliff AP and lead-gain differences versus ChemProp remain inconclusive.

Sources: frozen Moonshot reference metrics; `results/dablander_mmp_5fold/performance_summary.json`.

Impact, business model and next steps

Potential impact (if further validated)

More efficient analogue prioritization, with explicit uncertainty, to guide follow-up experiments.

Prospective users (not confirmed partners)

- Academic drug-discovery groups and medicinal chemistry teams
- Biotechnology start-ups and pharmaceutical R&D
- Research collaborations, possibly CRO partners

Possible sustainability models (proposals)

- Open research toolkit
- Research collaborations
- Optional hosted workflows / institutional integration

Business model (proposal)

Proposed: enterprise licenses for lead optimization or funded pharma co-development. Demand is untested.

Next-stage roadmap

✓ **Today: working prototype**

2 **DiffSBDD analogue generation**

3 **Structure-based scoring**

4 **Stereochemistry and 3D-aware evaluation**

5 **Expanded cliff validation**

6 **Experimental testing**
validate computational results in the lab